Total No. of Questions : 6]

SEAT No. :

P45

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APR - 18/TE/Insem. - 47

T.E. (Information Technology)

SYSTEMS PROGRAMMING

(2012 Course) (Semester - II) (314450)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) *Attempt Q1 or Q2, Q3 or Q4, and Q5 or Q6.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right indicate full marks.*
4) *Assume suitable data, if necessary.*

Q1) For the following assembly language program show MNT, MDT, Stack Organization and the expanded assembly language program **[10]**

```
        MACRO
        EVAL &X,&Y,&Z
        AIF (&Y EQ &Z) .ONLY
        MOVER AREG, &X
        SUB AREG,&Y
        ADD AREG,&Z
        AGO.OVER
.ONLY   MOVER AREG, &Z
.OVER   MEND
        MACRO
        ABC & MAIN
        MACRO
        & MAIN & W
        EVAL 1,1,3
        ADD AREG,&W
        EVAL 1,2,3
        MEND
        START 200
        ABC HELLO
        MOVEM  AREG, = '5'
        HELLO XYZ
        END.
```

OR

P.T.O.

Q2) a) Perform single pass assembler processing and generate symbol table, literal table, pool table, table of incomplete instructions(TOII) and target code for the given example. **[6]**

```
        START
        MOVER AREG, = '10'
        MOVEM BREG, SYM
        ADD AREG, BREG
        PRINT LOOP
SYM     MOVEM CREG, = '4'
        LTORG
        MOVER AREG, LOOP
LOOP    STORE BREG, = '3'
        END
```

Note : AREG and BREG are both registers.

b) With the structure List down the tables and functions of Pass I and Pass II of Macroprocessor. **[4]**

Q3) a) List and Explain basic functions of loader and explain how they are performed in absolute loader scheme. **[6]**

b) What is meant by Overlay structure in loaders? Discuss with examples. **[4]**

OR

Q4) a) Explain the working of compile and Go loader and absolute loader and compare the two. **[6]**

b) What information does the assembler provide the loader in case of the BSS loader. **[4]**

Q5) a) Convert the given regular expression to DFA: **[8]**

$((a + ba)^*(b + \varepsilon)) + b^*$

b) Define lexical analysis and syntax analysis phase of compiler. **[2]**

OR

Q6) a) Process the given statement w.r.t the first three phases of compiler. **[6]**

R = X + 10/(Y/Z ^ W)

Note : ^ is exponentiation operator.

b) Explain the structure of a LEX file. **[4]**

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